

Question	Answer	Marks
5(a)	energy transferred per unit charge (from electrical to other forms)	B1
5(b)(i)	heater	A1
5(b)(ii)	$(V =) 7.0 \times 0.86 = 6.0 \text{ (V)}$	A1
5(b)(iii)	$I = (230 - 6.0) / 170$ $(= 1.3 \text{ A})$	C1
	$I_1 = 7.0 - 1.3$ $= 5.7 \text{ A}$	A1
5(b)(iv)	$V = 230 - 6.0 - (5.7 \times 2.4)$	C1
	$= 210 \text{ V}$	A1
	or	
	$R = ((230 - 6.0) / 5.7) - 2.4$ $(= 36.9 \Omega)$	(C1)
	$V = 5.7 \times 36.9$ $= 210 \text{ V}$	(A1)

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5(b)(v)	$P = IV$ or $P = I^2R$ or $P = V^2 / R$	C1
	$P = 5.7 \times 210$ or $P = 5.7^2 \times (210 / 5.7)$ or $P = 210^2 / (210 / 5.7)$ $P = 1200 \text{ W}$	A1
5(b)(vi)	% efficiency = <u>useful</u> power out / (total) power in ($\times 100$)	C1
	$= 1200 / (230 \times 7.0) (\times 100)$ $= 0.75 (\times 100)$ $= 75\% \text{ or } 74\% \text{ (using 3 s.f. value from (v) gives 74\%)}$	A1
5(b)(vii)	(current) decreases	A1

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